

JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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ABSTRACT

Counterfactual explanations (CE) characterize how targeted input changes redirect a model’s prediction, yet prevailing generators routinely perturb far more features than the outcome requires, eroding the clarity and actionability on which the usefulness of such explanations depends. We introduce *COLA*, a post-hoc framework that is agnostic to both the predictive model and the CE generator and that refines any supplied counterfactual by computing an optimal-transport (OT) coupling between the factual and counterfactual sets and using that coupling to drive a Shapley-based attribution, *p*-SHAP, which isolates a minimal set of edits while preserving the target counterfactual effect. Beyond this algorithmic contribution, *p*-SHAP furnishes a unifying account of Shapley-style contrastive attribution: the widely used Baseline, Random-Baseline, and Counterfactual variants all arise as special cases under particular factual–counterfactual couplings. Two results delineate the framework’s guarantees: the OT coupling minimizes a transport-cost upper bound on the W_1 divergence between factual and counterfactual outcomes, and, under mild conditions, the refined counterfactual is provably no farther from the factual instance than the aligned reference it is built from. Across four tabular binary-classification datasets, twelve predictive models, and five CE generators spanning instance-based, group-based, and distributional paradigms, COLA reproduces the same target effects while retaining only 26–45% of the original feature edits, and on a tractable benchmark with known factual–counterfactual alignment it closely tracks the exact combinatorial optimum. Together these findings establish *p*-SHAP-guided refinement as a broadly applicable and dependable route to sparse, faithful counterfactual explanations.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) is a prerequisite for deploying artificial intelligence systems that are transparent and trustworthy (Arrieta et al., 2020; Das & Rad, 2020). Two complementary families of techniques dominate this landscape. Feature Attributions (FA) methods, most prominently Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017), quantify the contribution of each input feature to a machine learning (ML) model’s output and thereby distill a complex decision function into its most influential factors; in a clinical setting, for instance, Shapley values surface salient drivers such as age and medical history and so help practitioners interpret a model’s decisions (Ter-Minassian et al., 2023; Nohara et al., 2022). Counterfactual Explanations (CE) (Wachter et al., 2017; Guidotti, 2022), by contrast, articulate how targeted changes to input features would redirect the model toward a different outcome. Although hundreds of CE algorithms have now been proposed (Guidotti, 2022; Verma et al., 2020), no single algorithm serves every use case, precisely because each is deliberately tailored to a particular scenario, objective, and task. The governing objective varies accordingly: some algorithms seek a single counterfactual instance for each factual instance, whereas others treat multiple instances jointly as a group and return one or several counterfactuals per factual observation. A further class operates at the level of the entire dataset, identifying global CE that prescribe the direction along which factual instances should move to attain the desired model output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024); others recast the group of factual instances as an empirical distribution and search for a counterfactual distribution that preserves the shape of the factual distribution at comparable cost (You et al., 2025). The methods diverge,

finally, in their modeling assumptions: some presuppose differentiable models, while others are engineered specifically for tree-based or ensemble predictors. This heterogeneity is not incidental but structural, and it is exactly what motivates a refinement procedure that presumes neither a particular generator nor a particular model class.

Problem Description and Challenges Although FA and CE are each indispensable for rendering AI models interpretable and accountable, either technique used in isolation is deficient: FA alone rarely yields actionable steps, whereas CE alone tends to prescribe superfluous feature changes that are impractical to enact. We therefore address a general problem that couples the two, building on the extensive foundations established in the literature (see Appendix A). Concretely, we seek to answer the following question:

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$			$\mathbf{z}'$			$\mathbf{z}''$			$\mathbf{y}^*$
$\textcircled{a}$	$\textcircled{b}$	$\textcircled{c}$	$\textcircled{a}$	$\textcircled{b}$	$\textcircled{c}$	$\textcircled{a}$	$\textcircled{b}$	$\textcircled{c}$	$\textcircled{c}$
200	5	No	250	8	Yes	200	5	No	Yes
150	3	No	150	3	No	150	7	Yes	Yes
100	2	No	350	9	Yes	100	2	No	Yes
150	6	No	150	6	No	350	6	Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] A platform seeks to increase user registrations, having recorded interaction features such as the amount of money spent ($\textcircled{a}$), the number of clicks ($\textcircled{b}$), and registration status ($\textcircled{c}$). In the original data ($\mathbf{x}$), no user is registered. Two action plans, $\mathbf{z}'$ and $\mathbf{z}''$, each adjust user characteristics toward the desired outcome ($\mathbf{y}^*$) of full registration. Both attain the same (half) counterfactual effect, yet $\mathbf{z}''$ realizes it with strictly fewer modifications than $\mathbf{z}'$. This contrast crystallizes the central objective pursued throughout the paper: attaining a prescribed target effect at the minimal editing cost.

Addressing this problem entails overcoming three central challenges. First, because the problem is inherently task-specific, no single CE algorithm can be expected to satisfy every requirement, so a viable solution must operate on top of whichever generator is appropriate rather than replacing it. Second, to remain applicable across the full spectrum of predictive models, the approach must forgo strong structural assumptions such as differentiability or a prescribed model form. Third, and most subtly, generic FA—feature importance chief among them—can be actively misleading in this setting, because attribution scores computed without reference to the counterfactual need not cohere with the feature changes that actually realize the counterfactual effect; it is therefore ineffective to perform FA independently of CE to select the most important features to change. We demonstrate later that this decoupling induces counterproductive feature modifications (equivalently, the actions taken by users or stakeholders) in Result II of Section 6, precisely because the features deemed important in general do not coincide with the specific pathways that realize the desired counterfactual outcomes.

Main Contributions This paper advances four contributions, each substantiated by the developments that follow. (i) *Framework*. We introduce COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a general post-hoc framework that refines CE uniformly across predictive models and CE generators by using an optimal transport (OT)-induced coupling between the factual and counterfactual sets to guide Shapley attributions, thereby consolidating attribution and edit selection into a single modular pipeline whose four components can be interchanged independently. (ii) *Attribution method*. The framework yields p -SHAP, a joint-distribution-informed Shapley method that subsumes the commonly used Baseline, Random-Baseline, and Counterfactual Shapley variants as special cases under appropriate couplings, thereby exhibiting them as instances of one attribution principle rather than as disparate heuristics. (iii) *Theory*. We establish two supporting guarantees: the OT coupling minimizes a transport-cost upper bound on the W_1 divergence between factual and counterfactual outcomes, and, under mild conditions, the refined counterfactual is provably no farther from the factual instance than the aligned reference from which it is constructed. We are explicit that these results characterize the OT alignment and the proximity of the refinement rather than certifying approximation ratios for the underlying combinatorial selection problem, which is $\mathcal{NP}$ -hard. (iv) *Evidence*. Across four tabular binary-classification datasets, twelve predictive models, and five CE methods, COLA with

p -SHAP reproduces the same target effects while retaining only 26–45% of the original feature edits, and on a tractable benchmark with a known factual–counterfactual alignment it closely tracks the exact combinatorial optimum.

To the best of our knowledge, COLA is the first method to address the minimal-action problem of Figure 1 systematically *without committing to a particular CE algorithm or model class*. This agnosticism is not merely a convenience: it is the structural property that enables the broad applicability across models and generators that our experiments demonstrate.

2 PROBLEM FORMULATION

We now formalize the problem depicted in Figure 1. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ denote a black-box ML model. Denote by $\mathbf{x}$ the observed (factual) data comprising n rows and d columns ($\mathbf{x} \in \mathbb{R}^{n \times d}$, $n \geq 1$, $d \geq 1$), and let $\mathbf{y}^*$ denote the target model output ($\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$). The task is to determine a (group of) counterfactual instance(s) $\mathbf{z}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$) that drives the model output(s) as close as possible to $\mathbf{y}^*$ while sanctioning at most C feature changes. Writing D for a divergence that quantifies the dissimilarity between two entities, we state the problem as the following constrained program.

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) together with the constraint equation (1b) encodes the canonical CE optimization: $\mathbf{z}$ is driven so that $f(\mathbf{z})$ approaches $\mathbf{y}^*$ while remaining proximate to $\mathbf{x}$, so that $\mathbf{z}$ functions as a contrastive reference explaining why $f(\mathbf{x})$ fails to reach $\mathbf{y}^*$. We deliberately leave D unrestricted so as to preserve the generality of the formulation; admissible choices include the Euclidean distance, OT, maximum mean discrepancy (MMD), and mean or median differences of the model outcome. The remaining constraints equation (1c)–equation (1e) superimpose the action budget onto this optimization. Beyond the counterfactual data $\mathbf{z}$, we jointly optimize the binary indicator matrix $\mathbf{c}$, in which z_{ik} is permitted to change if and only if $c_{ik} = 1$; constraint equation (1c) caps the total number of such changes at C . Constraints equation (1d) and equation (1e) operationalize this coupling through a standard big- M mechanism: when $c_{ik} = 0$ they force $z_{ik} = x_{ik}$, leaving position (i, k) untouched, whereas when $c_{ik} = 1$ the sufficiently large constant M slackens the bounds and frees z_{ik} to vary.

To solve equation (1), we resort to FA to identify the most influential features to obtain the modification indicator variable $\mathbf{c}$. The next section introduces commonly used Shapley value methods for FA, which, together with our later proposed one, are integrated into our algorithmic framework COLA, to obtain the refined counterfactual $\mathbf{z}$. This attribution-guided strategy is warranted by the intrinsic hardness of the exact program: the problem is computationally intractable even in its simplest regime, being $\mathcal{NP}$ -hard already when $d = 1$ and f is linear (Appendix B). Because this precludes a scalable exact solver, a principled heuristic that concentrates the limited action budget on the most consequential edits is not merely convenient but necessary, and it is such a heuristic that the remainder of the paper develops.

3 PRELIMINARIES ON SHAPLEY VALUE

This section reviews the commonly used Shapley value methods for FA that, together with the method we later propose, are unified within the COLA framework. We first recall the Shapley value from cooperative game theory and then examine its principal instantiations for FA; Sundararajan & Najmi (2020) provide a comprehensive survey of these variants.

In cooperative game theory, a coalitional game is specified by a finite set of players—features, in our context—denoted $\mathcal{F} = \{1, 2, \dots, d\}$, together with a characteristic function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ that assigns to each coalition $S \subseteq \mathcal{F}$ the total payoff $v(S)$ its members can secure through cooperation, normalized so that $v(\emptyset) = 0$. The Shapley value, defined in terms of v below, is the canonical solution

concept in this setting: it furnishes the unique equitable allocation of the grand-coalition payoff $v(\mathcal{F})$ among the individual players.

Before defining the Shapley value, let $\Delta(k, S)$ denote the marginal contribution of a player (feature) k to a coalition S , that is, the additional payoff the coalition realizes upon including k :

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i for a player i in a cooperative game is defined in equation (3) below.

$$\phi_k = \frac{1}{d} \sum_{S \subset \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

The value ϕ_k thus averages the normalized marginal contributions $\Delta(k, S)$ of feature k over all coalitions S that exclude k , each weighted by the inverse binomial coefficient $\binom{d-1}{|S|}$, with d the total number of features. Summing over all such coalitions and dividing by d yields the axiomatically fair distribution of the total payoff among features, each term registering the contribution of the corresponding feature within the full collection.

For an arbitrary data point $\mathbf{x}_i$, we write $\mathbf{x}_{i,S}$ for its restriction to the features in S and $\mathbf{x}_{i,\mathcal{F} \setminus S}$ for the complementary part, applying the analogous convention to other notations. We now instantiate this framework in the three Shapley value methods most widely used for FA, each of which differs only in how it models the absence of a feature.

Baseline Shapley (B-SHAP). This method fixes a single baseline point $\mathbf{r}_j$ as the reference for every $\mathbf{x}_i$ and defines the characteristic function

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j,\mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

so that the absence of a feature is modeled by substituting its value from the baseline point $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). It thereby presupposes an exact, deterministic alignment between each factual instance in $\mathbf{x}$ and its reference in $\mathbf{r}$.

Random Baseline Shapley (RB-SHAP). This variant of B-SHAP, employed implicitly in Lundberg & Lee (2017) and in the widely used SHAP library, replaces the fixed baseline with an expectation over a background distribution:

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

Here the background distribution $\mathcal{D}$ is conventionally instantiated as the training dataset (Lundberg & Lee, 2017; Merrick & Taly, 2020), so that the reference is drawn marginally rather than aligned to any specific factual counterpart—a choice that, as our experiments show, discards precisely the alignment information the counterfactual problem hinges on.

Counterfactual Shapley (CF-SHAP). This method extends RB-SHAP by instantiating the background distribution $\mathcal{D}$ as a counterfactual distribution conditioned on each individual factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$):

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

Adopted in Albini et al. (2022); Kommiya Mothilal et al. (2021) with demonstrated advantages for contrastive FA, this method replaces the deterministic alignment of B-SHAP with a probabilistic one between $\mathbf{x}$ and $\mathbf{r}$, yet it inherits that alignment from the CE generation mechanism rather than optimizing it—a dependence our formulation removes.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We unify the family spanning equation (4) through equation (6) by parameterizing the reference through an algorithm A_{Prob} that returns the joint probability of the factual and counterfactual sets, whereupon the three baselines above emerge as special cases under particular

couplings.¹ Rather than treating these estimators as unrelated recipes, this parameterization exposes them as points along a single axis—the choice of factual–counterfactual coupling—and thereby clarifies exactly what each prior method assumes. Formally, p -SHAP is defined as follows,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_i, S; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

Crucially, whereas prior methods rest on random or heuristically fixed baselines, p -SHAP confronts the alignment problem directly, identifying the coupling through OT rather than inheriting it. This reframes contrastive feature attribution not as a bare decomposition of the prediction but as a transport problem that explicitly minimizes the cost of moving factual instances onto their counterfactual references—a conceptual shift with which the method’s empirical gains are, as Section 6 documents, consistently associated.

We emphasize that A_{Prob} is independent of the CE algorithm and depends only on the generated counterfactual $\mathbf{r}$. Because it conditions solely on the factual set and the returned counterfactual, p -SHAP yields attributions invariant to the idiosyncrasies of the underlying CE generation process—a deliberate departure from CF-SHAP, whose reference distribution remains tethered to that process. This invariance is what allows a single attribution routine to serve instance-based, group-based, and distributional generators alike. Contrary to common expectations, we demonstrate that OT can be more effective than relying on a counterfactual distribution defined by a CE generation mechanism as done by Albini et al. (2022), in Result II of Section 6 later.

The central instantiation studied in this paper takes A_{Prob} to be the OT problem (equivalently, the squared 2-Wasserstein divergence) defined below, whose optimal transportation plan $\mathbf{p}_{\text{OT}}$ then serves as the joint distribution of $\mathbf{x}$ and $\mathbf{r}$ within p -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Here $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ denote the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$, and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of couplings with these marginals, i.e. all admissible transport plans. The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is the entropic regularizer with coefficient $\varepsilon \geq 0$; activating it ($\varepsilon > 0$) renders the objective strictly convex and hence solvable by fast Sinkhorn iterations, which accelerate the computation of OT and, as detailed in the complexity analysis, keep the coupling step cheap relative to the generator itself.

Theoretical Aspects of p -SHAP By construction, the OT coupling minimizes the total feature-modification cost of moving $\mathbf{x}$ toward $\mathbf{r}$ under the alignment it induces between the factual set $\mathbf{x}$ and the counterfactual set $\mathbf{r}$ —precisely the criterion our problem demands, namely attaining the counterfactual outcome by adjusting $\mathbf{x}$ toward $\mathbf{r}$ at minimal editing cost. We formalize this correspondence under the assumption that the predictive model f is Lipschitz continuous. Theorem 4.1 below (proved in Appendix C) shows that the plan $\mathbf{p}_{\text{OT}}$ employed by p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$: the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$ is dominated by the Lipschitz constant times the square root of the minimized expected transport cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, whose entries p_{ij} ($j = 1, 2, \dots, m$) prescribe how the feature values of $\mathbf{x}_i$ should be adjusted toward those of one or several $\mathbf{r}_j$. The practical consequence is direct, if modest in scope: within p -SHAP, the OT plan $\mathbf{p}_{\text{OT}}$ supplies a modification strategy that steers $\mathbf{x}$ toward $\mathbf{r}$ at the smallest expected cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ attainable over all couplings $\mathbf{p} \in \Pi$. Because $\mathbf{p}_{\text{OT}}$ attains this minimal cost, it simultaneously furnishes the tightest bound of this form on the counterfactual-effect violation $W_1(f(\mathbf{x}), \mathbf{y}^*)$, linking the quality of the alignment to transport optimality. We stress that this is a statement about a Lipschitz-induced surrogate for the counterfactual effect, not an optimality certificate for the discrete edit-selection objective itself; it motivates the OT coupling as a principled proxy rather than establishing an approximation ratio for the final sparse solution.

¹First, p -SHAP degrades to B-SHAP in equation (4) when A_{Prob} defines a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates a deterministic $i \leftrightarrow j$ alignment for any $\mathbf{x}_i, \mathbf{r}_j$. Second, p -SHAP degrades to RB-SHAP in equation (5) when A_{Prob} is independent of CE and instead relies on an arbitrary background distribution $\mathcal{D}$. Third, p -SHAP degrades to CF-SHAP in equation (6) when A_{Prob} is built upon a known conditional distribution of CE.

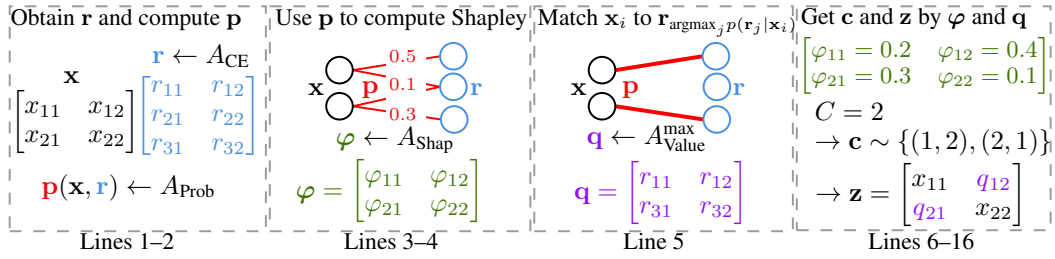


Figure 2: [An illustration of COLA] The figure traces how COLA derives $\mathbf{c}$ and $\mathbf{z}$ for equation (1). For clarity we adopt the simpler value-assignment variant A_{Value}^{max} in line 5. In lines 6–16 we set $C = 2$, so that sampling from the probability matrix φ designates exactly two positions for modification.

Theorem 4.1 (*p*-SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Although Theorem 4.1 bounds the transport cost rather than the discrete objective directly, that cost serves as a tractable convex proxy for the $\mathcal{NP}$ -hard sparsity (L_0) problem. By concentrating attribution mass on the most cost-efficient transport paths, the OT coupling steers the selection step in Algorithm 1 toward discarding non-essential features. Beyond this transport-cost argument, *p*-SHAP admits a principled interventional interpretation: it attributes the change in model behavior to modifications of the corresponding features, as formalized in Theorem 4.2 below (proved in Appendix D).

Theorem 4.2 (Interventional Effect of *p*-SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Finally, *p*-SHAP inherits the axiomatic guarantees of B-SHAP and RB-SHAP, which underpin its soundness as a feature-attribution tool; we omit the proof and refer the reader to Sundararajan & Najmi (2020); Lundberg & Lee (2017) for the Shapley axioms.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch The COLA framework, given in Algorithm 1, addresses equation (1) by composing four interchangeable algorithmic components, each responsible for one stage of the pipeline: a CE generator A_{CE} that produces the counterfactual, a coupling routine A_{Prob} that returns the joint distribution of the factual and counterfactual sets, a Shapley attributor A_{Shap} that scores each factual entry under that coupling, and a value assignment A_{Value} that supplies the target values for the selected edits. It is precisely the modularity of this decomposition—the fact that each stage exposes a fixed interface and can be replaced without altering the others—that renders COLA agnostic to the specific model and generator. We detail the four stages below, with an accompanying illustration in Figure 2.

Line 1 (Applying a CE algorithm to obtain a counterfactual $\mathbf{r}$). The CE generator A_{CE} receives the model f , the factual $\mathbf{x}$, the target outcome $\mathbf{y}^*$, and the tolerance ϵ , and returns a counterfactual $\mathbf{r}$ that

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

remains proximate to $\mathbf{x}$ while satisfying $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$. Any generator conforming to this interface is admissible, and it is exactly this minimal interface—rather than any assumption on how the counterfactual is produced—that renders the subsequent refinement generator-agnostic.

Line 2 (*Constructing a joint distribution of $\mathbf{x}$ and $\mathbf{r}$*). The routine A_{Prob} takes $\mathbf{x}$ and $\mathbf{r}$ and returns a matrix encoding the joint distribution over the n factual points of $\mathbf{x}$ and the m counterfactual points of $\mathbf{r}$. This joint distribution $\mathbf{p}$ specifies the alignment (matching) between the rows of $\mathbf{x}$ and those of $\mathbf{r}$ and is subsequently consumed in Line 5 to derive the values from which $\mathbf{z}$ is composed. As established in Section 3, when the row-wise alignment is unknown—the prevailing case in practice— A_{Prob} is instantiated through OT, yielding the coupling that minimizes the OT distance between $\mathbf{x}$ and $\mathbf{r}$; when a faithful alignment is instead known a priori, the corresponding joint distribution can be supplied directly.

Lines 3–4 (*p-SHAP FA*). We evaluate equation (7) to compute the Shapley values of $\mathbf{x}$. The joint distribution $\mathbf{p}$ enters at this stage to align each row of $\mathbf{x}$ with its counterfactual rows in $\mathbf{r}$, so that the selected rows of $\mathbf{r}$ serve as the most representative contrastive reference for that row of $\mathbf{x}$; this alignment is the decisive determinant of the resulting contrastive FA, a claim our ablations in Section 6 bear out directly. The Shapley-value matrix of $\mathbf{x}$ is then taken element-wise in absolute value and normalized to sum to one, and the resulting matrix φ constitutes our attribution.

Line 5 (*Computing feature values*). The routine A_{Value} takes the counterfactual $\mathbf{r}$ and the joint distribution $\mathbf{p}$ and, for each factual row i of $\mathbf{x}$, selects one or several rows of $\mathbf{r}$ to reference when editing $\mathbf{x}$. It returns a matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$ whose entry q_{ik} furnishes the counterfactual candidate value for x_{ik} ($\forall i, k$). We instantiate two variants, $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, corresponding respectively to selecting a single row and to aggregating multiple rows.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\text{max}}$ selects the single row of $\mathbf{r}$ carrying the highest coupling probability.

$$\mathbf{q} = A_{\text{Value}}^{\text{max}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i), k} \text{ and } \tau(i) = \arg \max_{j=1, 2, \dots, m} p_{ij}. \quad (9)$$

The alternative $A_{\text{Value}}^{\text{avg}}$ instead forms q_{ik} as a convex combination, namely a coupling-weighted average of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Because the non-negative matrix φ is normalized to sum to one, it can be read as a sampling policy over edit positions: line 7 draws the positions of $\mathbf{x}$ designated for replacement, setting $c_{ik} = 1$. For every (i, k) with $c_{ik} = 1$, the entry x_{ik} is reset to q_{ik} ; the resulting matrix is returned as $\mathbf{z}$, which together with $\mathbf{c}$ constitutes the solution of equation (1).

By Theorem 4.1, COLA is designed to minimize the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ by modifying $\mathbf{z}$ based on feature attribution results, which identify the most important features to adjust to achieve the desired counterfactual effect. The theorem below (proved in Appendix E) establishes a complementary proximity guarantee: the refined $\mathbf{z}$ produced by COLA provably satisfies the proximity constraint equation (1b) in the canonical setting $n = m$ under the Frobenius norm, so that refinement never increases the distance to the factual instance. Table 3 corroborates empirically that this guarantee persists across the broader settings we evaluate.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Corollary 5.2. *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Complexity of COLA Let $\mathcal{O}(M_{\text{CE}})$ denote the complexity of the generator A_{CE} , and let M_{Shap} denote the number of sampled subsets when A_{Shap} estimates Shapley values by weighted linear regression—the practical approximation employed throughout our experiments, since exact Shapley computation is exponential in d . The overall complexity of COLA, as a function of n, m, d , and the entropic regularization parameter ε , decomposes additively across its four stages as $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where the value-assignment term is $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ under $A_{\text{Value}}^{\text{max}}$ and $N = \mathcal{O}(nmd)$ under $A_{\text{Value}}^{\text{avg}}$. Two properties follow. First, the attribution cost scales linearly in the feature dimension d for a fixed sampling budget M_{Shap} , so the refinement remains tractable as d grows. Second, the refinement overhead incurred beyond generating the counterfactual is polynomial in the problem dimensions, whence COLA imposes only a modest cost on top of any chosen CE generator; Appendix F provides the full derivation.

6 NUMERICAL RESULTS

This section evaluates COLA on the problem in equation (1), with $\mathbf{y}^* = f(\mathbf{r})$ for the counterfactual $\mathbf{r}$ returned by a CE generator A_{CE} . To probe robustness across divergence choices, we instantiate D with four distinct functions: OT measures the distance between full distributions; MMD measures the divergence between distribution means in a high-dimensional feature space; and absolute mean difference (MeanD) and absolute median difference (MedianD) measure the divergence in mean and median, respectively. The experiments are organized to substantiate three claims, each addressed by a dedicated result below: I) *COLA attains substantial modification minimality across settings.* II) *p-SHAP systematically outperforms the other Shapley methods on modification minimality.* III) *COLA closely tracks the exact optimum on tractable benchmarks.*

Experiment Setup We evaluate over a broad benchmark spanning 4 tabular binary-classification datasets, 5 CE algorithms chosen to span distinct design goals, and 12 classifiers, summarized in Table 1. Each combination of a dataset, a generator A_{CE} , and a model f constitutes an “experiment scenario.” Deliberately varying all three axes is what permits us to assess COLA independently of any particular model class or generator, and thereby to attribute its behavior to the refinement procedure rather than to a favorable pairing.

The generators A_{CE} in Table 1 were selected to span the three established CE paradigms. DiCE and KNN are instance-based, producing counterfactual(s) for each individual factual instance; AReS and GlobeCE are group-based, producing a collection of counterfactuals for the factual data as a whole;

Table 1: *[Experiment Scenarios Setup]* Four tabular binary-classification datasets benchmark five CE algorithms across 12 predictive models. A “scenario” denotes one combination of a dataset, a generator A_{CE} , and a model f ; the full cross-product of these three axes defines the evaluation space over which COLA is assessed.

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: *[Experiment Methods Setup]* The 6 compared methods, color-coded to match Figure 3. Each method is fully determined by a specific choice of the coupling A_{Prob} and the Shapley routine A_{Shap} , and each is benchmarked within the scenarios of Table 1. The proposed method is CF- p_{OT} , which realizes p -SHAP; the remaining rows are ablation baselines constructed so that each pairwise comparison isolates the contribution of a single component.

Method	The probability $\mathbf{p}$ used by A_{Value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{Prob}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{Prob}: \text{Eq. equation (8) (but not used in } A_{Shap})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{Prob}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{Prob}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{Prob}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{Prob}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

and Discount is distributional, treating the factual instances as an empirical distribution and seeking a proximate counterfactual distribution. Covering all three paradigms is what lets us test the central claim that COLA refines counterfactuals uniformly, irrespective of how they were generated.

Table 2 specifies the 6 methods, with CF- p_{OT} realizing the proposed p -SHAP and the remaining five serving as baselines; each is benchmarked comprehensively across the scenarios of Table 1. A method is fully determined by its triple of components A_{Prob} , A_{Value} , and A_{Shap} . For instance, RB- p_{Uni} takes A_{Prob} to be the uniform distribution when computing $\mathbf{p}$, passes the resulting $\mathbf{p}$ to A_{Value} via equation (9) or equation (10) to obtain $\mathbf{q}$, and sets A_{Shap} to RB-SHAP.

We emphasize that this suite is engineered as a controlled ablation, in which each pairwise comparison varies exactly one component while holding the others fixed, so that any performance difference is attributable to that component alone. First, RB- p_{Uni} and CF- p_{Uni} differ only in that the latter incorporates CE information whereas the former does not, isolating the value of using CE at all. Second, CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all inject CE into FA but under different joint distributions, isolating the effect of the coupling and, as shown below, exposing a pronounced advantage for the OT coupling. Third, contrasting CF- p_{OT} , which exploits the OT alignment, with RB- p_{OT} , which does not (the two differing only in A_{Shap}), confirms that OT helps precisely by informing A_{Shap} about the factual-counterfactual alignment rather than through any incidental confound. Finally, CF- p_{Ect} constitutes the idealized case in which each counterfactual has a known source, supplying an exact factual-counterfactual alignment and collapsing CF-SHAP onto B-SHAP. This design permits sharp internal attribution of the gains; it does not, however, benchmark against external sparse- or actionable-counterfactual optimizers, a comparison we identify as a natural direction for future work.

Result I: COLA delivers substantial action reduction at negligible cost in counterfactual effect
In Table 3, we require the refined $\mathbf{z}$ to reach either 80% or 100% of the counterfactual effect of $f(\mathbf{r})$,

Table 3: [COLA for Modification Minimality] For each method, the number of modified features in the refined counterfactual $\mathbf{z}$ required to attain the 80% and 100% counterfactual effect of $f(\mathbf{r})$, reported alongside the corresponding relative edit magnitude of $\mathbf{z}$. Each entry averages 4 scenarios drawn at random from Table 1, with 100 runs each; “-” marks scenarios in which the target counterfactual effect is unattainable by the method. Bold entries denote the best (smallest) value in each dataset block; across all four datasets and both effect levels, every such entry is achieved by the proposed CF- p_{OT} , and CF- p_{OT} is moreover the only method that reaches the full 100% effect on any dataset.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	5.29(± 0.09)	75.9%	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	—	—	—	—
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	—	—
	RB- p_{OT}	8.55(± 0.08)	50.2%	—	—
	CF- p_{Uni}	7.01(± 0.07)	41.1%	—	—
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	—	—
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	2.80(± 0.04)	34.4%	—	—
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	—	—
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	—	—
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

so that $f(\mathbf{z})$ recovers the target model output². COLA meets this requirement with markedly fewer actions in the refined $\mathbf{z}$ —that is, fewer feature modifications in $\mathbf{z}$ —than the original CE $\mathbf{r}$ carries. Reaching the 80% counterfactual effect demands only 13%–25% of the original feature changes, as measured by the relative edit magnitude $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$. More telling still, COLA with p -SHAP (CF- p_{OT}) is the sole method that attains the full 100% counterfactual effect on any dataset, and it does so using merely 26%–45% of the feature changes present in $\mathbf{r}$. This dominance is uniform: in every dataset block CF- p_{OT} records the lowest modification count and the smallest relative edit magnitude, which establishes p -SHAP as the decisive driver of action minimality within this comparison. We isolate the source of this advantage next.

Result II: p -SHAP systematically outperforms the other Shapley methods in achieving the counterfactual effect Figure 3 evaluates the Shapley methods of equation (4)–equation (7), plotting the residual divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ from equation (1) against the action budget C . Three findings emerge, each traceable to a single controlled contrast. First, RB- p_{Uni} and RB- p_{OT} trail the remaining methods by a wide margin, confirming that CE information is indispensable for attributing features toward modification minimality. Second, the fact that CF- p_{OT} surpasses RB- p_{OT} —the two differing only in A_{Shap} (Table 2)—pinpoints the mechanism of the gain: OT benefits p -SHAP specifically by supplying an effective factual–counterfactual alignment, not through any incidental factor. Third, p -SHAP decisively outperforms CF- p_{Uni} and CF- p_{Rnd} , which quantifies the value of the OT joint distribution in particular: merely injecting counterfactual information into FA, as CF- p_{Uni} and CF- p_{Rnd} do, is insufficient, and a *proper* alignment is required—one that, as Figure 4 reveals, need not coincide with the alignment dictated by the exact counterfactual generation mechanism.

²By definition $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, which represents a 100% counterfactual effect since $\mathbf{y}^* = f(\mathbf{r})$. A counterfactual effect of 80% is defined through the proportion of divergence eliminated by the refined CE $\mathbf{z}$, i.e. $Counterfactual\ Effect = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D being OT.

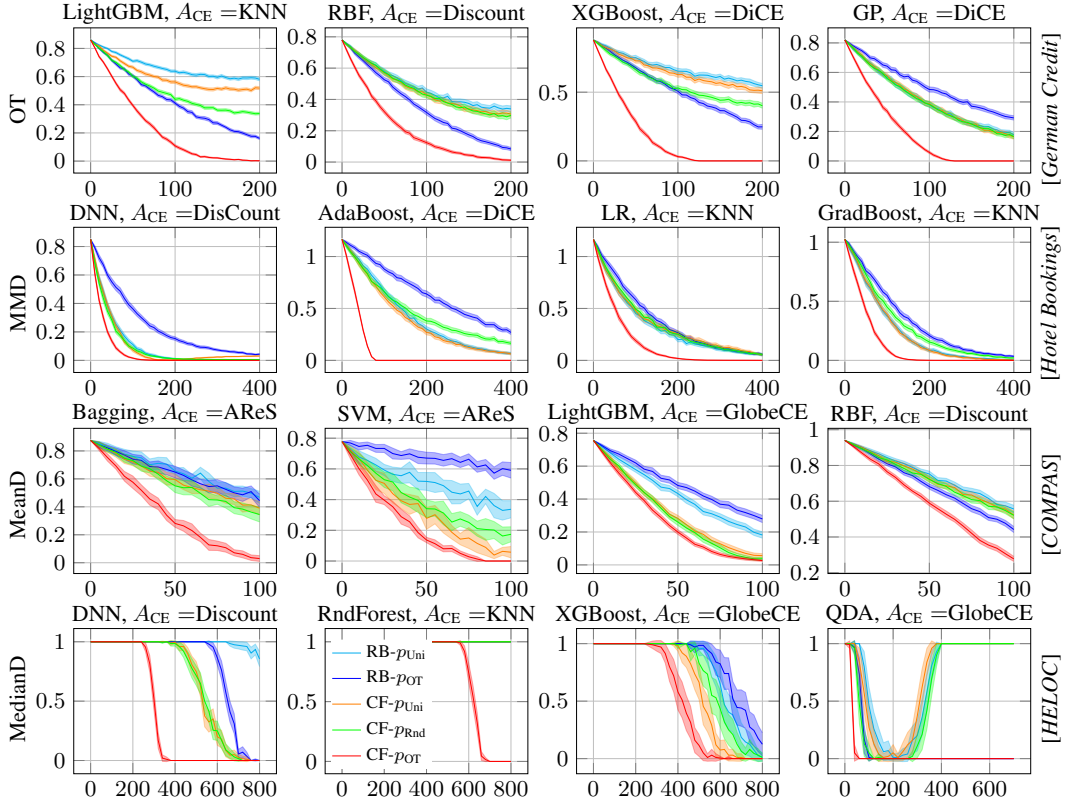


Figure 3: Residual divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C , averaged over 100 runs; shaded bands denote 99.9% confidence intervals. $A_{\text{Value}}^{\text{avg}}$ is used for *HELOC* and *COMPAS*, and $A_{\text{Value}}^{\text{max}}$ for *German Credit* and *Hotel Bookings*; the legend in the “RnDForest, $A_{\text{CE}} = \text{KNN}$ ” panel applies to all plots. Uniformly across datasets and models, CF- p_{OT} (red) drives the divergence down fastest and reaches the target effect at the smallest action budget C , with its advantage over the competing couplings widening as the residual divergence approaches zero.

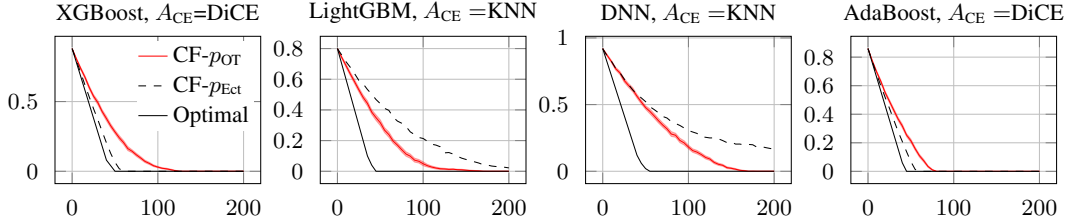


Figure 4: [*German Credit*] Residual divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ versus the action budget C , with D the *MeanD*. The mixed integer linear programming (MILP) solution furnishes the exact optimum—the lowest attainable divergence at each budget—against which the alignment-informed methods CF- p_{OT} and CF- p_{Ect} are compared. Both track the optimum closely, and CF- p_{OT} occasionally matches or surpasses the known-alignment variant CF- p_{Ect} .

This advantage is not confined to a handful of favorable settings. COLA with p -SHAP performs strongly across the experiment scenarios of Table 1, as evidenced jointly by Figure 3 and Table 3 and corroborated by the extensive further results reported in Appendix H. Within the suite of Shapley couplings we compare, then, p -SHAP is the consistent leader on counterfactual sparsity.

Result III: COLA is competitive with the MILP optimum on tractable benchmarks This result demonstrates the effectiveness of p -SHAP in eliminating the influence of the CE generation process

by replacing the CE algorithm-dependent knowledge³ of $\mathcal{D}$ with the OT joint distribution between the factual and counterfactual data, shown in Figure 4. We benchmark CF- p_{Ect} within COLA on equation (1) using MeanD as the divergence D . Because CF- p_{Ect} presupposes a known factual-counterfactual alignment, this setting isolates how effectively COLA exploits such an alignment when it happens to be available. The exact optimum for this case is obtained by solving an MILP, whose formulation is derived in Appendix G; because that MILP is computationally expensive, we solve it only for *German Credit*, which affords a tractable ground-truth reference. This benchmark therefore substantiates the optimality claim in a controlled, small-scale regime rather than at the full scale of the main experiments—a boundary we state plainly.

Within this regime, CF- p_{Ect} attains near-optimal performance under DiCE. Since the factual-counterfactual pairs are mutually independent for DiCE and KNN, this establishes that COLA is effective for instance-based CE, even though the formulation equation (1) is stated generally enough to encompass group and distributional CE. Notably, CF- p_{OT} occasionally surpasses even CF- p_{Ect} , because it utilizes a more theoretically grounded approach to identify the key features that require modification, whereas CF- p_{Ect} relies on CE algorithm-dependent knowledge, which lacks solid justification on its effectiveness for FA. That the OT coupling can outperform an exact known alignment underscores our central thesis: a well-chosen coupling, rather than the generator’s own alignment, is what drives sparse attribution. A residual gap between CF- p_{Ect} , CF- p_{OT} , and the optimum persists under KNN, so identifying the best attainable alignment remains an open problem.

7 CONCLUSIONS

This paper introduced COLA, a framework agnostic to both the predictive model and the CE generator that refines counterfactual explanations through joint-distribution-informed Shapley values, preserving the target counterfactual effect while sharply curtailing the number of required actions. The framework rests on two supporting guarantees—an OT coupling that minimizes a transport-cost upper bound on the W_1 divergence, and a proximity bound ensuring the refined counterfactual is no farther from the factual than the aligned reference—which motivate the design without purporting to certify the $\mathcal{NP}$ -hard combinatorial selection directly. Validated across four tabular binary-classification datasets, twelve predictive models, and five generators spanning the instance-based, group-based, and distributional paradigms, COLA with p -SHAP consistently reproduces the intended effect at 26%–45% of the original edits and closely tracks the exact optimum wherever it can be computed. Beyond these empirical gains, p -SHAP contributes a unifying account under which Baseline, Random-Baseline, and Counterfactual Shapley are recovered as special couplings. Natural extensions include benchmarking against external actionable-counterfactual optimizers, incorporating feasibility and immutability constraints into the selection step, and scaling to higher-dimensional feature spaces; within the tabular setting studied here, the evidence positions joint-distribution-informed attribution as a broadly applicable and dependable route to sparse, faithful counterfactual explanations.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ This paper proposes a method (COLA with p -SHAP) for refining CE. The contribution is methodological and empirical: it engages neither human subjects nor interventions, and no new personal data were collected, annotated, or released.

All experiments rely exclusively on standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC), used solely for benchmarking in accordance with their licenses and terms and with common research practice. We attempt no re-identification or linkage and release no additional personal information; our implementation will expose options to exclude protected attributes and to enforce feasibility and immutability constraints on features.

CE can surface biases latent in the underlying models but do not by themselves guarantee fairness. Our method reduces the number of recommended feature edits; it neither authorizes edits to sensitive or

³This knowledge, i.e. exact alignment between factual and counterfactual, is available only by DiCE and KNN among all CE methods considered.

⁴<https://iclr.cc/public/CodeOfEthics>

immutable attributes (e.g., sex, race, age) nor confers causal validity. We accordingly discourage the use of CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals into unrealistic or unethical behavioral changes. Any deployment in consequential domains should incorporate domain-expert-defined feasibility constraints, fairness audits (e.g., disparate impact or benefit across groups), human oversight, and compliance with applicable data-protection and AI regulations. We stress that our attribution is distributional rather than causal and must not be construed as a causal effect.

The principal risks are threefold: (i) misinterpretation of attributions as causal, (ii) recommendation of infeasible or harmful edits, and (iii) facilitation of strategic manipulation. We mitigate them by (a) stating the non-causal scope and its limitations explicitly, (b) supporting cost/feasibility masks and immutable features in the implementation, (c) reporting performance across subgroups where relevant, and (d) furnishing guidance for responsible use in the documentation.

To facilitate replication, we provide an anonymized implementation together with detailed settings and seeds; we report model types, hyperparameters, and compute budgets, and we guard against selective reporting through fixed evaluation protocols. To our knowledge, no conflicts of interest or sensitive sponsorships bear on this submission, and any funding disclosures will be provided following the double-blind review.

REPRODUCIBILITY STATEMENT

We regard reproducibility as integral to the contribution and provide pointers to every component required to reproduce our results. The COLA framework and the p-SHAP attribution are fully specified in the main paper through pseudocode and notation (Algorithm 1; the Method and Preliminaries sections), while all theoretical assumptions and complete, line-by-line proofs of the theorems and lemmas are provided in the appendix.

Implementation details—hyperparameters, optimization settings, environment specifications, and random seeds—are documented in Appendix I, which additionally describes the datasets and the evaluation protocol, including data splits and preprocessing. The baselines (predictive models and CE generators) and coupling choices are enumerated in the experimental sections and ablations, and their configurations are mirrored in the supplemental materials. The experiment code and software will be released upon acceptance; Appendix I further reports hardware and runtime to contextualize compute requirements, and the appendix supplies the ablation and sensitivity studies together with the small-scale MILP benchmark setup, so that both the main comparisons and the optimality benchmark admit independent verification.

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A COMPARISON TO EXISTING APPROACHES

Albini et al. (2022) introduced CF-SHAP, which employs counterfactual data points as the background distribution for Shapley values. Its formulation, however, presumes that the counterfactual distribution is defined conditionally on each individual instance, thereby positing a known probabilistic alignment between every factual and every counterfactual instance; analogous assumptions underlie (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). This premise breaks down in the many settings that call for global explanations. We show, in contrast, that the counterfactual-distribution setup of (Albini et al., 2022) is a special case of ours, recovered when A_{Prob} is fixed to the generator’s own conditional alignment; p -SHAP thus generalizes CF-SHAP rather than competing with it. Kwon & Zou (2022) proposed WeightedSHAP, which weights features rather than treating their contributions as equal; our method instead weights the contributions of data points and extends naturally to weighting rows and columns jointly. Kommiya Mothilal et al. (2021) establishes a framework that couples FA and CE for explainability, but its CE-based FA inherits the same assumption as (Albini et al., 2022), which impedes generalization to group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025). A further related line is (Sharma et al.), which models counterfactuals through a system’s causal structure and time-series predictors to decompose a single observed change in a system-level metric fairly into per-input contributions. Crucially, none of these works addresses the minimal-action CE problem that motivates the present paper.

Closest in spirit to our objective are the formulations of (Kanamori et al., 2022; Karimi et al., 2021). Karimi et al. (2021) minimizes the cost of performing actions under the assumption of a known structural causal model (SCM)—knowledge rarely available in practice, and a requirement COLA deliberately avoids. Kanamori et al. (2022) observe that it remains open whether existing CE methods can be repurposed to solve that problem, and propose an MILP-based approach for linear classifiers, tree ensembles, and deep ReLU networks, building on (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). Solving such an MILP is costly and scales poorly, whereas COLA attains sparse edits through a polynomial-time attribution step atop any generator. These optimization-based actionable-counterfactual methods constitute the most natural external baselines for the minimal-action problem, and a direct empirical comparison against them—beyond the internal ablations reported here—is a concrete and worthwhile direction for future work.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 below establishes that no scalable exact algorithm for solving it in general can be expected unless $\mathcal{P} = \mathcal{NP}$. The hardness stems not merely from the potential non-linearity of f but, more fundamentally, from the combinatorial structure of the edit-selection problem itself, which persists even for the simplest linear models.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. We proceed by reduction from the *Sparse Regression (SR) with a Cardinality Constraint* problem, defined as follows. Given a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the task is to find a vector $\mathbf{z} \in \mathbb{R}^n$ that minimizes the residual error subject to carrying at most K non-zero elements:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ denotes the number of non-zero elements in $\mathbf{z}$. This problem is $\mathcal{NP}$ -hard precisely because selecting which subset of variables to admit into the model is combinatorial.

We now map this SR instance onto our problem equation (1) through the following identifications. Set the number of features to $d = 1$ and the number of instances to n (matching the dimension of the SR problem). Take the factual data to be the zero vector $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$ and the target model output to be $\mathbf{y}^* \in \mathbb{R}^m$ as given in the SR problem, and define the model f as the linear map $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

For the model output we adopt the squared Euclidean distance $D(f(\mathbf{z}), \mathbf{y}^*) = \|f(\mathbf{z}) - \mathbf{y}^*\|_2^2$, and for the instances $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, the second equality following from $\mathbf{x} = \mathbf{0}$. The action budget C is set equal to the sparsity level k of the SR problem, and the big- M constant may be any sufficiently large positive number, for instance $M \geq \max_i |z_i|$.

Because $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) reduce to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

Hence $c_i = 0$ forces $z_i \leq 0$ and $z_i \geq 0$ simultaneously, so $z_i = 0$, whereas $c_i = 1$ admits $z_i \in [-M, M]$. Constraint equation (1c) correspondingly becomes $\sum_{k=1}^n c_i \leq k$.

Under these identifications, our problem equation (1) reduces to:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this formulation the binary variables c_i encode whether z_i is permitted to change ($c_i = 1$) or is pinned to zero ($c_i = 0$), so the constraint $z_i = 0$ whenever $c_i = 0$ mirrors the support constraint of the SR problem, and $\sum_{k=1}^n c_i \leq k$ enforces the same cardinality bound. The objective is, by construction, identical to that of SR.

Because this reduction places an $\mathcal{NP}$ -hard cardinality-constrained SR instance inside our problem, solving equation (1) is at least as hard as solving SR. Consequently, equation (1) is $\mathcal{NP}$ -hard even in the restricted regime where $d = 1$, the model f is linear, and the divergence D is the standard Euclidean distance. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the set of factual data points with associated probability weights $\mu_i \geq 0$ such that $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the set of counterfactual data points with associated probability weights $\nu_j \geq 0$ such that $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\mu, \nu)} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below shows that, under this coupling, the feature attributions align with the expected costs of feature modification, so that the induced action plans are cost-efficient in achieving the counterfactual outcome.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\boldsymbol{\pi} \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Proof. Since the model f is Lipschitz continuous with constant L , for any $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$, it holds that:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides of the inequality by $p_{ij}^{OT} \geq 0$, we obtain:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing both sides over all $i = 1, \dots, n$ and $j = 1, \dots, m$, we have:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let us denote $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. The inequality then becomes:

$$E_f \leq L E_d.$$

To further bound E_d , we apply the Cauchy-Schwarz inequality. Observe that the weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$ because $\mathbf{p}_{OT}$ is a probability distribution over the joint space of $\mathbf{x}$ and $\mathbf{r}$. The Cauchy-Schwarz inequality states that for any real-valued functions a_{ij} and b_{ij} ,

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy-Schwarz inequality:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the inequality for E_f , we have:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f^5 , then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Therefore, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimizes the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$, we have

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

Hence the conclusion follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ be the joint probability of $\mathbf{x}$ and $\mathbf{r}$ obtained from A_{Prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are set to $\mathbf{x}_{i,S}$, and the features in $\mathcal{F} \setminus S$ remain distributed according to their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Therefore, the expected value of $f(\mathbf{r})$ under the intervention is:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

Remark that by the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Hence the conclusion. $\square$

The value function $v^{(i)}(S)$ thus captures the expected value of $f(\mathbf{r})$ under the intervention that sets the features in S to $\mathbf{x}_{i,S}$, denoted $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. Since this intervention does not depend on any predefined joint distribution $p(\mathbf{x}, \mathbf{r})$, the expression $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ decomposes cleanly into the base expectation of $f(\mathbf{r})$ plus the additional interventional impact registered by the attribution $v^{(i)}(S)$.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{value}}(\mathbf{r}, \mathbf{p})$. For the elements where $c_{ik} = 1$ (modified elements), $z_{ik} = q_{ik}$. For elements where $c_{ik} = 0$, $z_{ik} = x_{ik}$. Therefore, we can write:

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms with respect to $\mathbf{q}$ and $\mathbf{z}$ are computed as follows:

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds because of equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, it follows that:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots, we get:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

Hence the conclusion. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Proof. Under the assumptions $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ becomes a permutation matrix associated with a bijection σ . Consequently, the value assignment algorithm A_{value} (whether maximizing or averaging) assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting $\mathbf{q}$ with the permuted matrix $\mathbf{r}_\sigma$ into the general bound established in Theorem 5.1 (i.e., $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$) directly yields the result. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We analyze the computational complexity of COLA stage by stage.

Consider first A_{Prob} . When the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, A_{Prob} merely materializes the matrix $\mathbf{p}$ from that prior knowledge, at cost $\mathcal{O}(n \times m)$. Otherwise, solving the OT problem to obtain $\mathbf{p}$ via the Sinkhorn–Knopp algorithm costs $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Consider next A_{Shap} . Each sampled subset requires evaluating the model on all n data points, incurring $\mathcal{O}(n)$ evaluations per subset. Incorporating the baseline values from the reference data $\mathbf{r}$ amounts to overwriting selected features with their counterparts drawn from the reference table $\mathbf{r}$; accessed naively this costs $\mathcal{O}(m)$, but when the reference values are precomputed and retrievable in constant time the per-subset cost reduces to $\mathcal{O}(d)$ across the d features. Aggregating over the M_{Shap} sampled subsets yields M_{Shap} model evaluations, so the overall cost of A_{Shap} is $\mathcal{O}(n \times d \times M_{\text{Shap}})$, linear in the feature dimension d for a fixed sampling budget.

The normalization in line 4 of COLA costs $\mathcal{O}(n \times d)$.

To compare the two value-assignment routines $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, we account for their per-row costs. Under $A_{\text{Value}}^{\text{max}}$, each row $\mathbf{x}_i$ requires: (1) computing the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, at $\mathcal{O}(m)$ per row; (2) identifying the reference row $\mathbf{r}_j$ of maximal probability p_{ij} , again $\mathcal{O}(m)$ per row; and (3) the assignment $q_{ik} = r_{\tau(i),k}$ with $\tau(i) = \arg \max_j p_{ij}$, at $\mathcal{O}(d)$ per row. Over n rows this totals $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

Under $A_{\text{Value}}^{\text{avg}}$, each row $\mathbf{x}_i$ requires: (1) computing the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, at $\mathcal{O}(m)$ per row; (2) forming the normalizer $\sum_{j'=1}^m p_{ij'}$, also $\mathcal{O}(m)$ per row; and (3) evaluating the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, at $\mathcal{O}(m \times d)$ per row. Over n rows this totals $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the aggregate cost is $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$, since $C \leq n \times d$.

Therefore, the complexity of COLA using $A_{\text{Value}}^{\text{max}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA using $A_{\text{Value}}^{\text{avg}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence the overall complexity of COLA in terms of n , m , d , and the entropic-OT regularization parameter ε is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

This section derives a global-optimality benchmark for the case of a known factual–counterfactual alignment when solving equation (1) with D instantiated as MeanD, namely

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Because COLA guarantees $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$ together with $\mathbf{z}$ remaining at least as close to $\mathbf{x}$ as $\mathbf{r}$ (Theorem 5.1), the proximity constraint equation (1b) is automatically satisfied and

may be dropped. The formulation of equation (1) then specializes to:

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

The original constraints equation (1d) and equation (1e) collapse into the single equality equation (13c) precisely because CF- p_{Ect} is imposed: for every $\mathbf{x}_i$ there exists an exact reference $\mathbf{r}_j$ serving in both A_{Shap} and A_{Value} , so each x_{ik} ($k = 1, 2, \dots, d$) either remains unchanged or is set to r_{jk} . The refined value $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$ is thus governed entirely by the binary variable c_{ik} .

Under this known alignment between each $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$, the reference matrix $\mathbf{q}$ is fully determined—and, notably, $A_{\text{Values}}^{\max}$ and $A_{\text{Values}}^{\text{avg}}$ return the identical $\mathbf{q}$. For any data point i and any feature set $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denote the solution $\mathbf{z}_i$ in which every feature $k \in S$ is changed to q_{ik} while each remaining feature $h \in \mathcal{F} \setminus S$ stays at x_{ih} . The family $\{\mathbf{z}_{iS}\}_{S \subseteq \mathcal{F}}$ therefore enumerates the entire domain of admissible values of $\mathbf{z}$. For each $\mathbf{z}_{iS}$ define the corresponding scalar

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ admits an exact representation through a binary selector a_{iS} paired with the scalar g_{iS} , namely,

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The optimization problem equation (13) is hence reformulated as the mixed-integer program below.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing η subject to equation (14b) and equation (14c) is equivalent to minimizing the objective of equation (13), since together these two constraints bound the absolute value from above by η . Constraint equation (14d) guarantees that each data point i is assigned exactly one feature-modification plan a_{iS} for some $S \subseteq \mathcal{F}$, and equation (14e) enforces the action budget of equation (13b). Solving equation (14) therefore yields the exact optimum of COLA under a known factual-counterfactual alignment, which serves as the ground-truth reference plotted in Figure 4 of Section 6.

H EXTENDED NUMERICAL RESULTS

Observing Figures 5–8, CF- p_{Uni} generally performs better than RB- p_{Uni} . Second, consider RB- p_{OT} and CF- p_{OT} that also differ only in A_{Shap} , the latter consistently outperforms the former. Hence *RB-SHAP is not suitable for FA in CE*.

We analyze how different Shapley methods affect FA, corresponding to lines 3–4 in COLA. The shapley methods can be classified into two categories: First, consider RB- p_{Uni} and CF- p_{Uni} that differ only in A_{Shap} . Observing Figures 5–8, CF- p_{Uni} generally performs better than RB- p_{Uni} . Second,

consider $RB-p_{OT}$ and $CF-p_{OT}$ that also differ only in A_{Shap} , the latter consistently outperforms the former. Hence *RB-SHAP is not suitable for FA in CE*.

Besides FA, the other equally important step of COLA is line 5, i.e. using the joint probability $p(\mathbf{x}, \mathbf{r})$ to compose the matrix $\mathbf{q}$, telling the factual x to which direction to change its features so as to move towards the target model outcome. We observe in Figures 5–8 that $CF-p_{OT}$ consistently outperforms all other methods throughout all experiments. Note that all the three methods $CF-p_{Uni}$, $CF-p_{Rnd}$, and $CF-p_{OT}$ provide solution’s for the joint probability $\mathbf{p}$ when the exact alignment between factials and counterfactuals are unknown. Yet, their performance differ significantly. Simply knowing the CE (and its marginal distribution) is insufficient.

OT proves to be exceptionally useful when the alignment information between factual and counterfactual instances is missing or inaccurate. Even when the CE algorithm explicitly matches each factual instance to a corresponding counterfactual, it is challenging to justify that the known alignment optimizes performance. This is supported by Figure 4 in Section 6.

Note that p_{OT} does not need to be the true joint distribution of $\mathbf{x}$ and $\mathbf{r}$ from a data generation perspective. Instead, it should guide COLA to treat $\mathbf{x}$ and $\mathbf{r}$ together for both FA and CE. Furthermore, the QDA column in Figure 5 shows stableness of OT-based methods, while others diverge significantly from the target. We emphasize that COLA, however, is *not limited to using OT as A_{Prob}* . As indicated by Figure 4, any known best $\mathbf{p}$ still has non-negligible gap to the global optimality. Devising a better A_{Prob} algorithm is hence an interesting topic worth exploration.

I EXPERIMENTS REPRODUCIBILITY

The experiments are conducted on a high performance computing (HPC) cluster, running with four nodes (for the four datasets) in parallel, with each node equipped with two Intel Xeon Processor 2660v3 (10 core, 2.60GHz) and 128 GB memory. The experiment runs approximately 5-10 hours in each node, dependent on the size of the dataset. It is also possible to reproduce the experiment on a laptop, while it costs more computational time generally than using an HPC cluster.

For the four datasets, the numerical features are standardized, and the categorical features follow either label-encoding or one-hot encoding. Practically, we did not observe remarkable difference between the two encoding methods in terms of COLA’s performance. The train-test split follows 7 : 3.

The optimality baseline as shown in Figure 4 is solved by Gurobi 11.0.2 ([gur](#)). In order to reproduce the optimality baseline, a license of Gurobi is required. Otherwise, we can resort to open-source operations research libraries such as Google-OR tools ([goo](#)). We remark that solving the MILP in Appendix G is computationally expensive, such that it may only apply to small scale datasets such as German Credit. If one wants to compute the optimality baseline for other datasets, then the number of used features needs to be reduced.

The hyperparameters of the models used in the experiment are specified as follows. The models Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA are scikit-learn models ([skl](#)), where all hyper-parameters are kept default. The models DNN, SVM, RBF, and LR are implemented by PyTorch ([Paszke et al., 2019](#)). The DNN has three layers. The SVM uses the linear kernel. The models XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are used by their scikit-learn interface, with all hyper-parameters kept default.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We used large language models (LLMs) only as general-purpose assist tools. Their role was limited to grammar, wording, and light copy-editing of author-written text. LLMs did not contribute research ideation, modeling choices, experimental design, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors. Any LLM-suggested phrasing was reviewed and edited before inclusion in the paper. The authors take full responsibility for all content; LLMs are not authors or contributors. This disclosure complies with the ICLR policy on LLM usage.

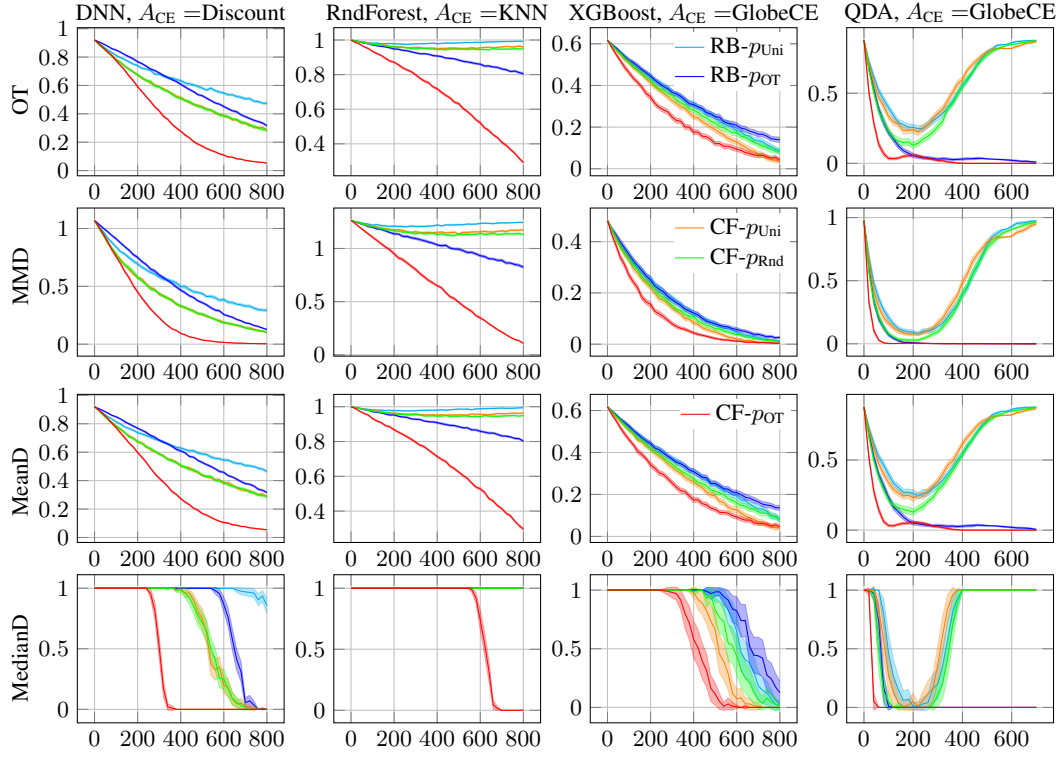


Figure 5: [HELOC] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

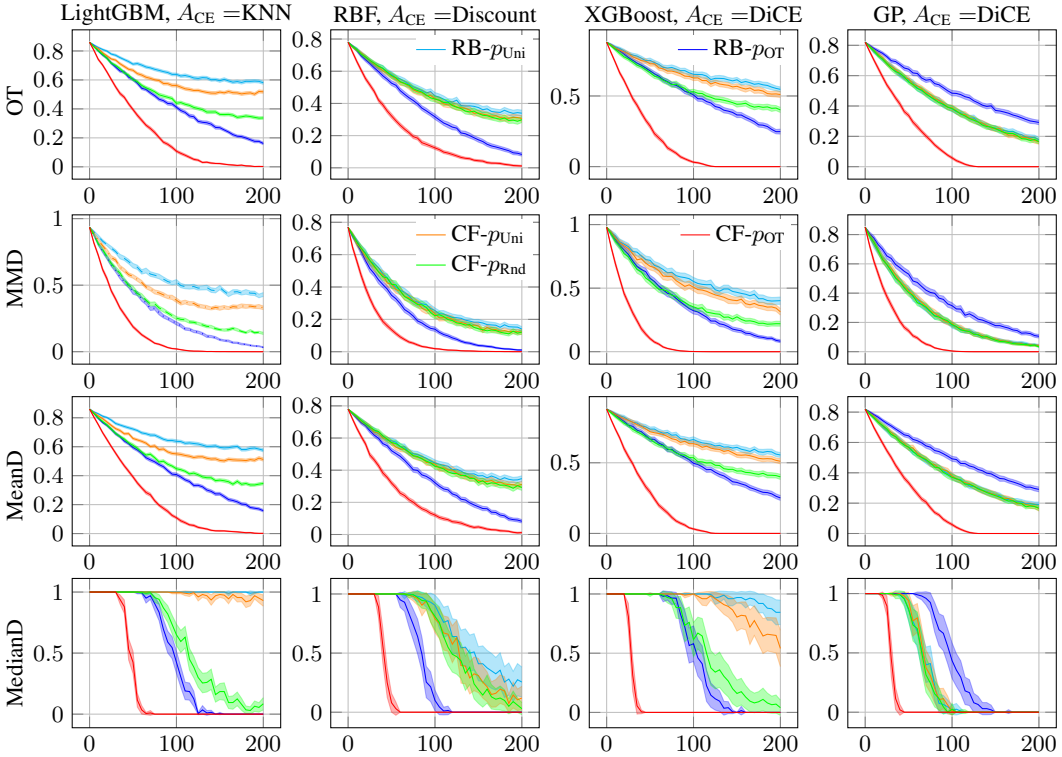


Figure 6: [German Credit] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

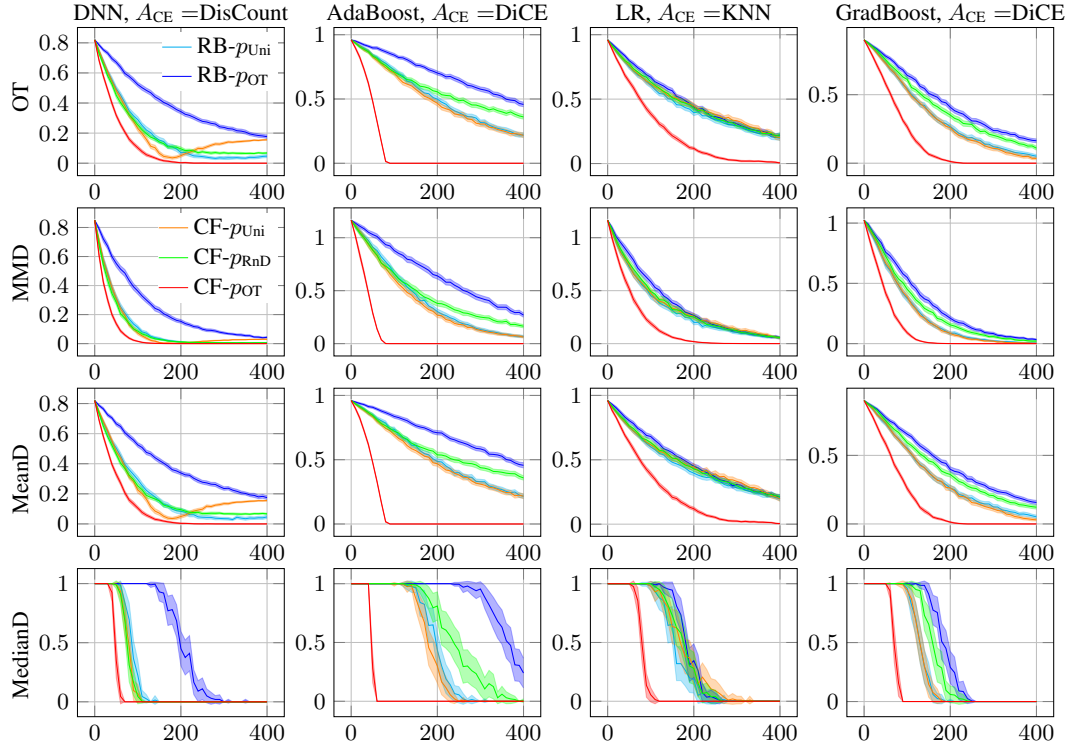


Figure 7: [Hotel Bookings] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.

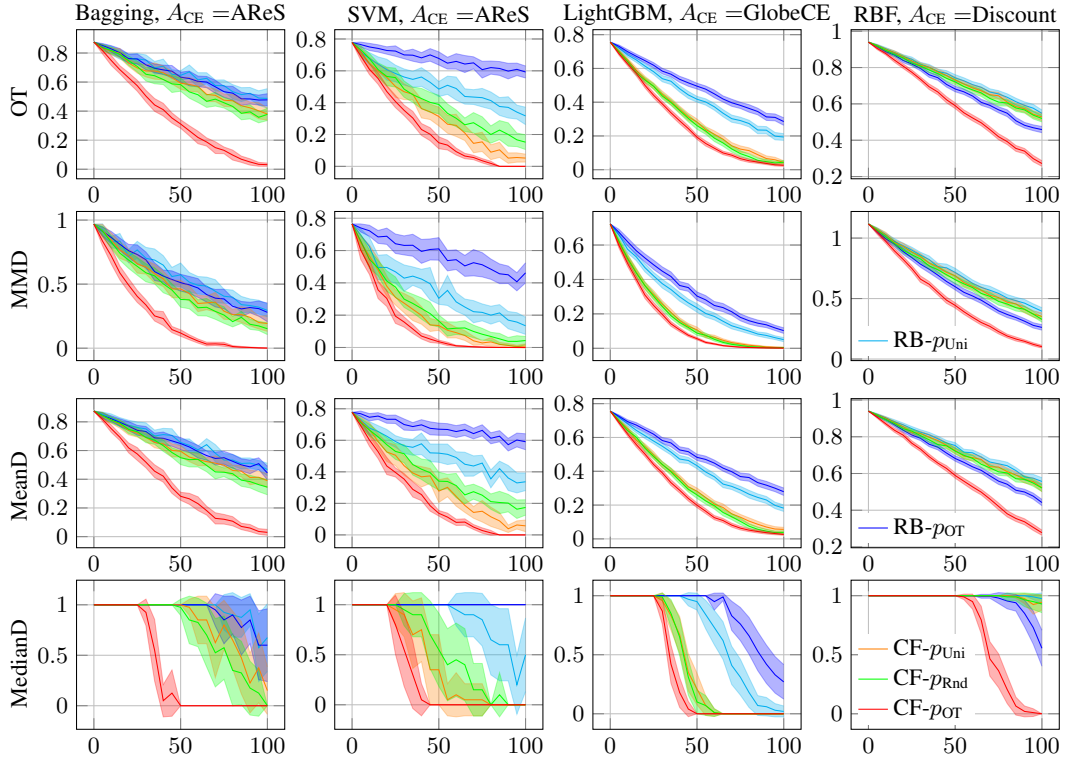


Figure 8: [COMPAS] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.